

Karshin Sachdeva

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EDUCATION

Georgia Institute of Technology, Atlanta, GA
Bachelor of Science in Computer Science

May 2028
GPA: 4.0/4.0

EXPERIENCE

OMRON Automation and Robotics

Software Engineering Intern (Computer Vision)

May 2026 – July 2026
Melbourne, Australia

- Deployed an automated inspection pipeline with FZ-PanDA that identifies defects on commercial products with a 95%+ detection accuracy across 8 different products running on live factory lines
- Implemented a multi-modal recognition system that combined OCR with color and shape criteria to distinguish between similar products running on a shared line, each triggering a dedicated defect inspection routine
- Linked vision system output with factory PLCs over EtherNet/IP which stored product pass/fail results in PostgreSQL and displayed defect trends alongside live operations on a real-time Grafana dashboard

Swinburne University of Technology – Factory of the Future

Undergraduate Research Assistant

June 2026 – Present
Melbourne, Australia

- Applied a crack detection pipeline for identifying defected concrete sleepers using YOLO v11 that ran an iterative reinforcement learning loop on a database of 9,000 images
- Generated synthetic defect imagery with a CycleGAN to reduce an imbalance in image counts across the 6 defect types and increasing detection precision and accuracy

Georgia Institute of Technology - Big Data Big Impact

Machine Learning Developer

Aug 2025 – Present
Atlanta, Georgia

- Developed a (2+1) D CNN that classified boxer's punches with PyTorch, NumPy and OpenCV on Google Colab for a full-stack boxing application with a classification accuracy of 91%
- Processed spatiotemporal data provided by Google MediaPipe with automated scripts for video data collection, which increased the training data supply by over 200% and standardized the data pipeline for pose sequence analysis

Georgia Institute of Technology - RoboJackets

Software Team Member

Aug 2025 – Present
Atlanta, Georgia

- Improved inter-robot communication using C and enhanced robot dribbling performance with real-time control logic
- Debugged ROS2-based software to allow for the use of RQT tools and robot controllers in virtual machine environments

PROJECTS

GeoGuessr Street View Geolocation Model

Jan 2026 - Present

- Built a 37,000-image dataset of Google Street View images by automatically sampling geographic coordinates for 37 European countries and rejecting positions that don't provide panoramic street view coverage
- Developed an EfficientNet-B4 classifier that is trained on a two-phase schedule and incorporates OCR-derived script supervision which recognizes 20+ languages and results in 36% accuracy which is 13x the random baseline

Pecho - Hackathon Project

Oct 2025 – Oct 2025

- Produced a voiced-based healthcare assistant with Twilio and OpenAI's API that answered medical questions, routed booking requests and updated personal records in a full stack application
- Implemented live conversation transcription with AssemblyAI so the agent's conversation can be dissected and responses tailored to resemble a medical professional

Music for the Hearing-Impaired

Mar 2021 – Feb 2025

- Engineered physical wristband prototype with the Raspberry Pi that converts WAV files into haptic vibrations through a motor controller and stepper motors in Python. Developed an accompanying iOS app using Swift and Xcode.
- Programmed the wristband's interface that was presented to 1000 active participants from the Auslan Deaf Community

Senior Citizen Mobility App

Mar 2024 – Dec 2024

- Programmed a webapp using HTML, CSS and JavaScript that implemented graphing algorithms like A*, Dijkstra and simulated annealing help senior citizens personalize their shopping experience when navigating malls

RELEVANT SKILLS

- **Software** – PyTorch, NumPy, Twilio, React, ROS2, RQT, Git, Google Maps API, FZ-PanDA, PostgreSQL, Grafana
- **Programming** - Java, Python, JavaScript, C/C++, x86 Assembly, Swift, HTML/CSS, IEC 61131-3
- **Languages** – English, Hindi
- **Relevant Coursework** – Object Oriented Programming, Data Structures & Algorithms, Linear Algebra, Computer Organisation and Programming, Discrete Math, Intro to Artificial Intelligence, Algorithm Design and Analysis